

Unable to obtain FakeAlmaden model

<https://github.com/Qiskit/qiskit-aer/issues/1192>

ROW 65

Unable to obtain FakeAlmaden model #1192

New issue

Closed



haoqindeng opened on Mar 24, 2021

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Informations

- Qiskit Aer version: 0.7.5
- Python version: 3.7.7
- Operating system: Windows Subsystem Linux 2

What is the current behavior?

We attempted to run the following code:

```
backend = FakeAlmaden()
fakealmaden_model = PulseSystemModel.from_backend(backend)
```



and we get the following error:

```
(vqe) haoqindeng@LAPTOP-CE39QN05: /mnt/c/Users/haoqi/OneDrive/Desktop/QCQ$ python qoc_test2.py line 45
Traceback (most recent call last):
  File "qoc_test2.py", line 46, in <module>
    fakealmaden_model = PulseSystemModel.from_backend(backend)
  File "/home/haoqindeng/miniconda3/envs/vqe/lib/python3.7/site-packages/qiskit/providers/aer/pulse/system_model.py", line 104, in from_backend
    return cls.from_config(config, subsystem_list)
  File "/home/haoqindeng/miniconda3/envs/vqe/lib/python3.7/site-packages/qiskit/providers/aer/pulse/system_model.py", line 114, in from_config
    hamiltonian = HamiltonianModel.from_dict(ham_string, subsystem_list)
  File "/home/haoqindeng/miniconda3/envs/vqe/lib/python3.7/site-packages/qiskit/providers/aer/pulse/system_model.py", line 124, in from_dict
    system.parse(subsystem_list)
  File "/home/haoqindeng/miniconda3/envs/vqe/lib/python3.7/site-packages/qiskit/providers/aer/pulse/system_model.py", line 134, in parse
    coef, token = self._tokenizer(ham, qubit_list)
  File "/home/haoqindeng/miniconda3/envs/vqe/lib/python3.7/site-packages/qiskit/providers/aer/pulse/system_model.py", line 144, in _tokenizer
    opr = gen_oper(name, idx, self.dim_osc, self.dim_qub)
  File "/home/haoqindeng/miniconda3/envs/vqe/lib/python3.7/site-packages/qiskit/providers/aer/pulse/system_model.py", line 154, in gen_oper
    return qobj.generators.tensor(opers)
  File "/home/haoqindeng/miniconda3/envs/vqe/lib/python3.7/site-packages/qiskit/providers/aer/pulse/qutip_extensions.py", line 104, in tensor
    return ten.tensor(list_qobj)
  File "/home/haoqindeng/miniconda3/envs/vqe/lib/python3.7/site-packages/qiskit/providers/aer/pulse/qutip_extensions.py", line 114, in tensor
    out.data = zcsr_kron(out.data, q.data)
  File "spmath.pyx", line 188, in qiskit.providers.aer.pulse.qutip_extensions.cy.spmath.zcsr_kron
  File "spmath.pyx", line 422, in qiskit.providers.aer.pulse.qutip_extensions.cy.spmath._safe_multiply
OverflowError: value too large
```



Steps to reproduce the problem

```
import qiskit
from qiskit import transpile, schedule as build_schedule
from qiskit.test.mock import FakeAlmaden
from qiskit import Aer, IBMQ, execute
from qiskit import QuantumCircuit
from qiskit.circuit import Gate

from qiskit import pulse
from qiskit.ignis.characterization.calibrations import rabi_schedules, RabiFitter

from qiskit.pulse import DriveChannel
from qiskit.compiler import assemble
from qiskit.qobj.utils import MeasLevel, MeasReturnType
# The pulse simulator
from qiskit.providers.aer import PulseSimulator

# Object for representing physical models
from qiskit.providers.aer.pulse import PulseSystemModel

# Mock Armonk backend
from qiskit.test.mock.backends.armonk.fake_armonk import FakeArmonk

# backend = qiskit.providers.aer.PulseSimulator()
backend = FakeAlmaden()
```



Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants



```

from qiskit.pulse import DriveChannel
from qiskit.compiler import assemble
from qiskit.qobj.utils import MeasLevel, MeasReturnType
# The pulse simulator
from qiskit.providers.aer import PulseSimulator

# Object for representing physical models
from qiskit.providers.aer.pulse import PulseSystemModel

# Mock Armonk backend
from qiskit.test.mock.backends.armonk.fake_armonk import FakeArmonk

# backend = qiskit.providers.aer.PulseSimulator()
backend = FakeAlmaden()
print(backend.configuration().hamiltonian)
# schedule = build_schedule(transpiled_circ, backend)
circ = QuantumCircuit(2, 2)
circ.x(0)
circ.x(0)
circ.x(1)
circ.measure([0, 1], [0, 1])

schedule = build_schedule(circ, backend,
                          method="as_late_as_possible")
# schedule.draw(channels=[pulse.DriveChannel(0), pulse.DriveChannel(1)])

backend_sim = PulseSimulator()

qobj = assemble(schedule,
                 backend=backend_sim,
                 meas_level=1,
                 meas_return='avg',
                 shots=1)

print('line 45')
fakealmaden_model = PulseSystemModel.from_backend(backend)
print('line47')
sim_result = backend_sim.run(qobj, fakealmaden_model).result()

```

What is the expected behavior?

We should be able to run the code at the line:
 fakealmaden_model = PulseSystemModel.from_backend(backend)
 instead of getting the error above.

Suggested solutions

Don't know how. Any ideas?

[@taalexander](#)



[haoqindeng](#) added **bug** on Mar 24, 2021



[nonhermitian](#) on Mar 24, 2021

Contributor ...

You cannot build a sparse representation of a system model that big. That is what the error is indicating.



[haoqindeng](#) on Mar 24, 2021

Author ...

Our ultimate goal is to get the last line "sim_result = backend_sim.run(qobj, fakealmaden_model).result()" to work, but we are currently struck at error at "fakealmaden_model = PulseSystemModel.from_backend(backend)". Basically we are trying to extract a schedule of a circuit "circ" with FakeAlmaden backend, then use this schedule to construct a qobj and run it with fakealmaden_model.

If we cannot build a model that big, what do you suggest we could do to execute the qobj and obtain the results?

[@nonhermitian](#)

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`fakealmden_model = PulseSystemModel.from_backend(backend)`
instead of getting the error above.

Suggested solutions

Don't know how. Any ideas?

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Our ultimate goal is to get the last line "`sim_result = backend_sim.run(qobj, fakealmden_model).result()`" to work, but we are currently struck at error at "`fakealmden_model = PulseSystemModel.from_backend(backend)`". **Basically we are trying to extract a schedule of a circuit "circ" with FakeAlmden backend, then use this schedule to construct a qobj and run it with fakealmden_model.**

If we cannot build a model that big, what do you suggest we could do to execute the qobj and obtain the results?

[@nonhermitian](#)



DanPuzzuoli on Mar 24, 2021

Contributor ...

Since it looks like you are only trying to run a schedule on the qubits 0 and 1, when constructing the `PulseSystemModel`, you can do:

```
fakealmden_model = PulseSystemModel.from_backend(backend, subsystem_list=[0, 1])
```



This will tell the constructor to only construct the model for just qubits 0 and 1.



chrisectlectic closed this as **completed** on Sep 15, 2021